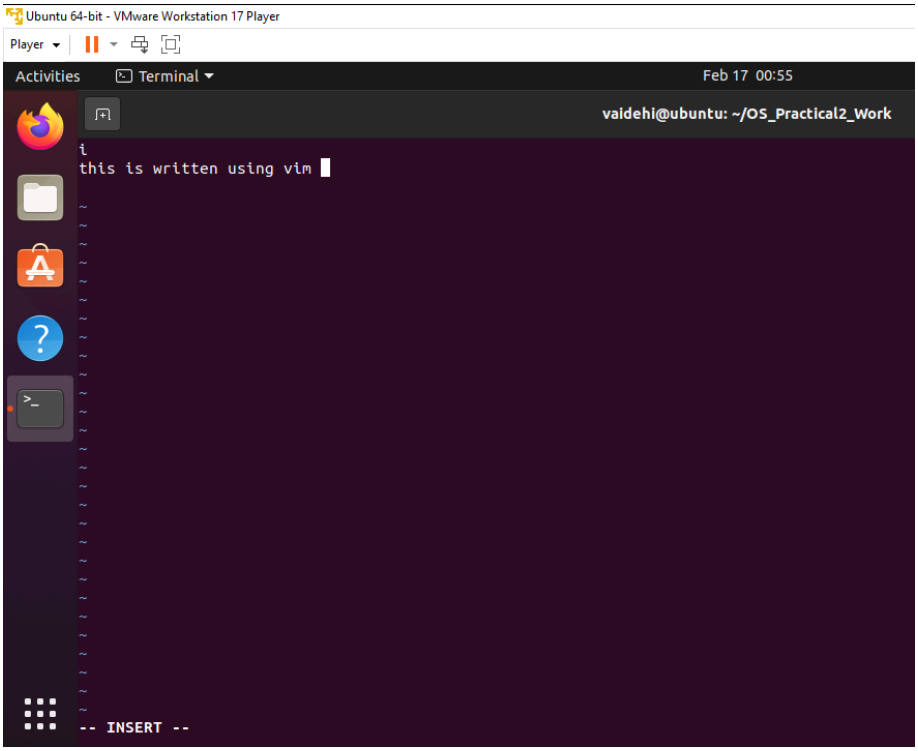
Activities Terminal Feb 17 00:35

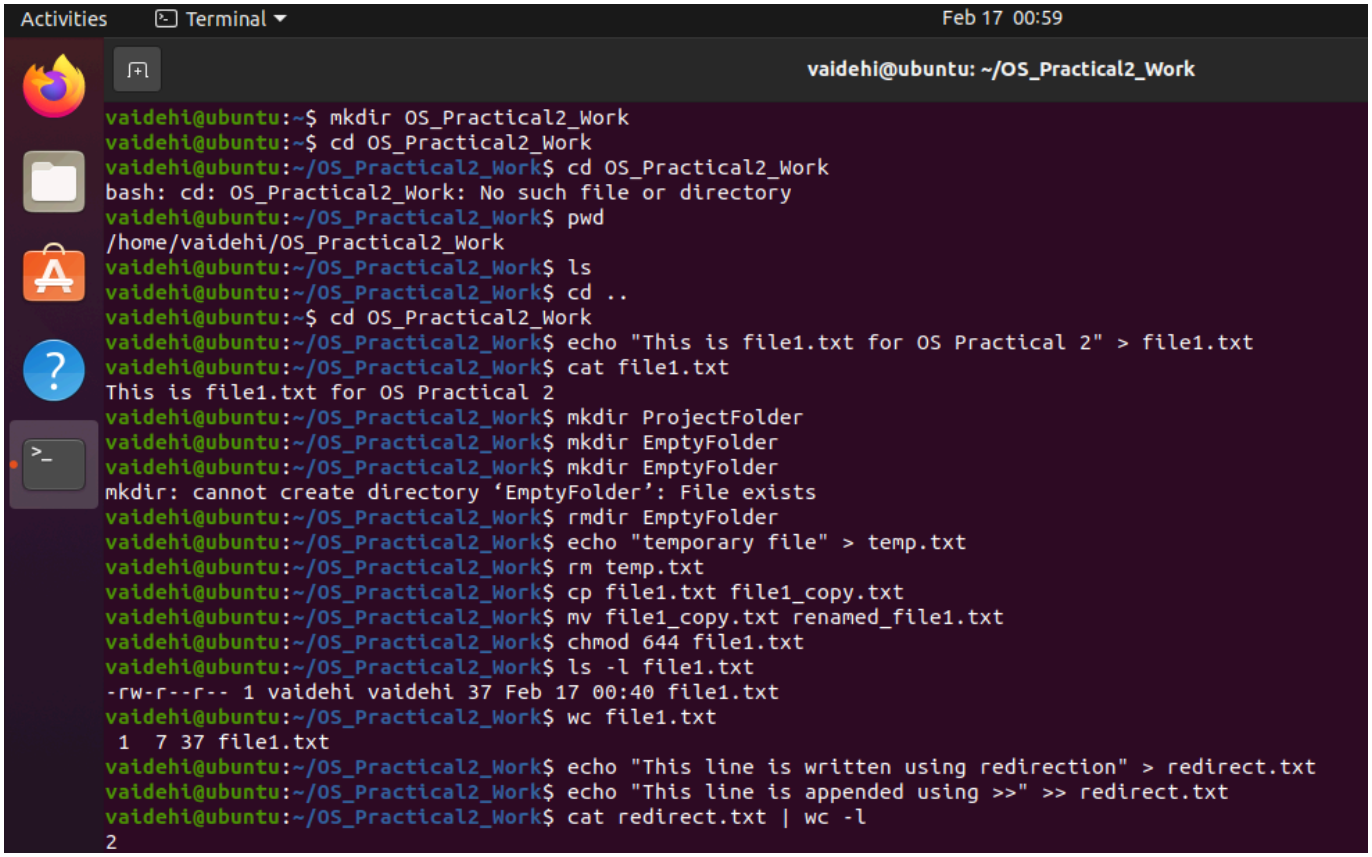
vaidehi@ubuntu: ~

```
vaidehi@ubuntu:~$ ls
Desktop     file_stat           fork_exec.c     syscalls          Vaidehi
Documents   file_stat.c         Music           syscalls.c        Videos
Downloads   file_stat.c.save    mypatch.patch   syscalls.c.save
file1.txt   file_stat.c.save.1  Pictures        Templates
file2.txt   fork_exec           Public          test.txt
vaidehi@ubuntu:~$ cd
vaidehi@ubuntu:~$ pwd
/home/vaidehi
vaidehi@ubuntu:~$ cat
^C
vaidehi@ubuntu:~$ cat stat.c
cat: stat.c: No such file or directory
vaidehi@ubuntu:~$ cat file_stat.c
#include <stdio.h>
#include <sys/stat.h>
#include <time.h>

int main(int argc, char *argv[])
{
    struct stat fileStat;

    // Check if filename is given
    if (argc < 2)
    {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }

    // Get file details
    if (stat(argv[1], &fileStat) < 0)
    {
        perror("stat");
```

vaidehi@

```
#include <time.h>

int main(int argc, char *argv[])
{
    struct stat fileStat;

    // Check if filename is given
    if (argc < 2)
    {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }

    // Get file details
    if (stat(argv[1], &fileStat) < 0)
    {
        perror("stat");
        return 1;
    }

    printf("File Name: %s\n", argv[1]);
    printf("File Size: %ld bytes\n", fileStat.st_size);

    printf("Number of Links: %ld\n", fileStat.st_nlink);
    printf("File Inode: %ld\n", fileStat.st_ino);

    printf("Last Access Time: %s", ctime(&fileStat.st_atime));
    printf("Last Modification Time: %s", ctime(&fileStat.st_mtime));
    printf("Last Status Change Time: %s", ctime(&fileStat.st_ctime));

    return 0;
}
vaidehi@ubuntu:~$ mkdir OS practical 2
```


Ubuntu 64-bit - VMware Workstation 17 Player
Player
Activities
Terminal
Feb 17 00:55
vaidehi@ubuntu: ~/OS_Practical2_Work
i
this is written using vim
-- INSERT --


Activities
Terminal
Feb 17 00:59
vaidehi@ubuntu: ~/OS_Practical2_Work
vaidehi@ubuntu:~$ mkdir OS_Practical2_Work
vaidehi@ubuntu:~$ cd OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ cd OS_Practical2_Work
bash: cd: OS_Practical2_Work: No such file or directory
vaidehi@ubuntu:~/OS_Practical2_Work$ pwd
/home/vaidehi/OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ ls
vaidehi@ubuntu:~/OS_Practical2_Work$ cd ..
vaidehi@ubuntu:~$ cd OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "This is file1.txt for OS Practical 2" > file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ cat file1.txt
This is file1.txt for OS Practical 2
vaidehi@ubuntu:~/OS_Practical2_Work$ mkdir ProjectFolder
vaidehi@ubuntu:~/OS_Practical2_Work$ mkdir EmptyFolder
vaidehi@ubuntu:~/OS_Practical2_Work$ mkdir EmptyFolder
mkdir: cannot create directory ‘EmptyFolder’: File exists
vaidehi@ubuntu:~/OS_Practical2_Work$ rmdir EmptyFolder
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "temporary file" > temp.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ rm temp.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ cp file1.txt file1_copy.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ mv file1_copy.txt renamed_file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ chmod 644 file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ ls -l file1.txt
-rw-r--r-- 1 vaidehi vaidehi 37 Feb 17 00:40 file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ wc file1.txt
1 7 37 file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "This line is written using redirection" > redirect.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "This line is appended using >>" >> redirect.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ cat redirect.txt | wc -l
2

```
vaidehi@ubuntu:~/OS_Practical2_Work$ grep "line" redirect.txt
This line is written using redirection
This line is appended using >>
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "linux practical" | tr 'a-z' 'A-Z'
LINUX PRACTICAL
vaidehi@ubuntu:~/OS_Practical2_Work$ echo -e "mango\napple\nbanana" > fruits.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ sort fruits.txt
apple
banana
mango
vaidehi@ubuntu:~/OS_Practical2_Work$ head -n 2 fruits.txt
mango
apple
vaidehi@ubuntu:~/OS_Practical2_Work$ tail -n 2 fruits.txt
apple
banana
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "Hello" > diff1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ echo "Hello World" > diff2.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ diff diff1.txt diff2.txt
1c1
< Hello
---
> Hello World
vaidehi@ubuntu:~/OS_Practical2_Work$ echo -e "apple\nbanana\nmango" > comm1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ echo -e "apple\nbanana\nmango" > comm1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ echo -e "banana\ngrapes\nmango" > comm2.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ sort comm1.txt > comm1_sorted.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ sort comm2.txt > comm2_sorted.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ comm comm1_sorted.txt comm2_sorted.txt
apple
                banana
        grapes
                mango
```

```
vaidehi@ubuntu:~/OS_Practical2_Work$ less fruits.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ more fruits.txt
mango
apple
banana
vaidehi@ubuntu:~/OS_Practical2_Work$ file fruits.txt
fruits.txt: ASCII text
vaidehi@ubuntu:~/OS_Practical2_Work$ type ls
ls is aliased to `ls --color=auto'
vaidehi@ubuntu:~/OS_Practical2_Work$ split -l 1 fruits.txt split_part_
vaidehi@ubuntu:~/OS_Practical2_Work$ cmp diff1.txt diff2.txt
diff1.txt diff2.txt differ: byte 6, line 1
vaidehi@ubuntu:~/OS_Practical2_Work$ mkdir TarFolder
vaidehi@ubuntu:~/OS_Practical2_Work$ cp fruits.txt TarFolder/
vaidehi@ubuntu:~/OS_Practical2_Work$ tar -cvf TarFolder.tar TarFolder
TarFolder/
TarFolder/fruits.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ tar -xvf TarFolder.tar
TarFolder/
TarFolder/fruits.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ find . -name "fruits.txt"
./fruits.txt
./TarFolder/fruits.txt
```

```
vaidehi@ubuntu: ~/OS_Practical2_Work
sudo apt install vim-nox     # version 2:8.1.2269-1ubuntu5.31
sudo apt install neovim      # version 0.4.3-3

vaidehi@ubuntu:~/OS_Practical2_Work$ sudo apt install vim
[sudo] password for vaidehi:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  vim-runtime
Suggested packages:
  ctags vim-doc vim-scripts
The following NEW packages will be installed:
  vim vim-runtime
0 upgraded, 2 newly installed, 0 to remove and 87 not upgraded.
Need to get 7,116 kB of archives.
After this operation, 34.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 vim-runtime all 2:8.1.2269-1ubun
Get:2 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 vim amd64 2:8.1.2269-1ubuntu5.32
Fetched 7,116 kB in 16s (455 kB/s)
Selecting previously unselected package vim-runtime.
(Reading database ... 163565 files and directories currently installed.)
Preparing to unpack .../vim-runtime_2%3a8.1.2269-1ubuntu5.32_all.deb ...
Adding 'diversion of /usr/share/vim/vim81/doc/help.txt to /usr/share/vim/vim81/doc/help.txt.vim-tin
Adding 'diversion of /usr/share/vim/vim81/doc/tags to /usr/share/vim/vim81/doc/tags.vim-tiny by vim
Unpacking vim-runtime (2:8.1.2269-1ubuntu5.32) ...
Selecting previously unselected package vim.
Preparing to unpack .../vim_2%3a8.1.2269-1ubuntu5.32_amd64.deb ...
Unpacking vim (2:8.1.2269-1ubuntu5.32) ...
Setting up vim-runtime (2:8.1.2269-1ubuntu5.32) ...
```

```
Activities   Terminal   Feb 17 01:02
vaidehi@ubuntu: ~/OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ vim vimfile.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ gzip fruits.txt
vaidehi@ubuntu:~/OS_Practical2_Work$ gunzip fruits.txt.gz
vaidehi@ubuntu:~/OS_Practical2_Work$ zip demo.zip file1.txt redirect.txt
  adding: file1.txt (stored 0%)
  adding: redirect.txt (deflated 21%)
vaidehi@ubuntu:~/OS_Practical2_Work$ unzip demo.zip
Archive:  demo.zip
replace file1.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
 extracting: file1.txt
replace redirect.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
vaidehi@ubuntu:~/OS_Practical2_Work$ sudo apt install mlocate -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  nocache
The following NEW packages will be installed:
  mlocate
0 upgraded, 1 newly installed, 0 to remove and 87 not upgraded.
Need to get 50.1 kB of archives.
After this operation, 258 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu focal/main amd64 mlocate amd64 0.26-3ubuntu3 [50.1 kB]
Fetched 50.1 kB in 1s (53.3 kB/s)
Selecting previously unselected package mlocate.
(Reading database ... 165401 files and directories currently installed.)
Preparing to unpack .../mlocate_0.26-3ubuntu3_amd64.deb ...
Unpacking mlocate (0.26-3ubuntu3) ...
Setting up mlocate (0.26-3ubuntu3) ...
update-alternatives: using /usr/bin/mlocate to provide /usr/bin/locate (locate) in auto mode
Adding group `mlocate' (GID 136) ...
Done.
Initializing mlocate database; this may take some time... done
```

```
Initializing mlocate database; this may take some time... done
Processing triggers for man-db (2.9.1-1) ...
vaidehi@ubuntu:~/OS_Practical2_Work$ sudo updatedb
vaidehi@ubuntu:~/OS_Practical2_Work$ locate file1.txt
/home/vaidehi/2/file1.txt
/home/vaidehi/OS_Practical2_Work/file1.txt
/home/vaidehi/OS_Practical2_Work/renamed_file1.txt
vaidehi@ubuntu:~/OS_Practical2_Work$
```

```
Activities   Terminal ▾                                   Feb 17  01:06
                                          vaidehi@ubuntu: ~/OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ pwd
/home/vaidehi/OS_Practical2_Work
vaidehi@ubuntu:~/OS_Practical2_Work$ mkdir ScenarioProject
vaidehi@ubuntu:~/OS_Practical2_Work$ cd ScenarioProject
vaidehi@ubuntu:~/OS_Practical2_Work/ScenarioProject$ echo "Scenario practical file" > scenario.txt
vaidehi@ubuntu:~/OS_Practical2_Work/ScenarioProject$ ls
scenario.txt
vaidehi@ubuntu:~/OS_Practical2_Work/ScenarioProject$ mv scenario.txt ..
vaidehi@ubuntu:~/OS_Practical2_Work/ScenarioProject$
vaidehi@ubuntu:~/OS_Practical2_Work/ScenarioProject$ cd ..
vaidehi@ubuntu:~/OS_Practical2_Work$ rmdir SenarioProject
rmdir: failed to remove 'SenarioProject': No such file or directory
vaidehi@ubuntu:~/OS_Practical2_Work$ rmdir ScenarioProject
vaidehi@ubuntu:~/OS_Practical2_Work$
```